

Variational methods and geometric emergence in Scalar-Angular Torsion Theory

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A covariant geometric field model, Scalar-Angular-Torsion (SAT), is analyzed under Lean 4 formal proof constraints, requiring explicit variational equations and robust conservation laws. The internal consistency of this structure is secured by **Lean 4 formal verification**, which enforces fundamental stability constraints, including Positive Effective Mass and Topological Dominance ($\mathbf{R} \ll \mathbf{1}$). The theory employs an underlying geometric ontology where particle worldlines (filaments) intersect a temporal sheet. The dynamics are derived from a hybrid action which must be σ -finite over the discrete twist ensemble (τ) and **Fréchet-differentiable** over the continuous field space. Varying the emergent metric yields the Emergent Einstein Equation, establishing the Micro-to-Macro Scaling Relations. Intrinsic quantum properties like Spin and Flavor are recovered via **Holonomy Quantization**. This framework operates with no numerical parameters introduced axiomatically, relying on a single dimensional anchor, the $\mathbf{Q} = \mathbf{1}$ effective mass ($\mathbf{m}_{\text{eff}} \approx 1.0073 \times 10^{-27}$ kg), which fixes the dimensionless coupling ratio $\mathbf{R} \approx 0.007304$, consistent with the fine-structure constant.

I. INTRODUCTION AND FORMAL RIGOR

The Scalar-Angular-Torsion (SAT) framework posits reality as a 4D network of worldlines (filaments) intersecting a temporal sheet (Σ_t), serving as a **mathematical model without ontological commitment**. Formal predictions are derived exclusively from the action principle and variational equations. The ultimate rigor of the SAT theory is secured by detailed formal analysis, which must be fully presented to support claims of completeness.

A. Formal Consistency and Dimensional Anchoring

The numerical predictions are supported by **Lean 4 formal certification**, which validates conservation laws and positivity constraints, such as **Positive Effective Mass** ($m_{\text{eff}} > 0$). Stable particle ground states are structurally constrained by the **UV Finiteness Lock** ($\mathbf{Q} \leq \mathbf{3}$). The framework introduces ****no numerical parameters axiomatically****. Scale is established by fixing the $Q = 1$ effective mass to $\mathbf{m}_{\text{eff}} \approx 1.0073 \times 10^{-27}$ kg, which structurally determines the dimensionless coupling ratio $\mathbf{R} \approx 0.007304$.

II. THE COVARIANT FORMALISM AND DYNAMICS

A. Total Covariant Action and Fields

The dynamics are derived from a **Total Covariant Action** (S_{SAT}). The action is composed of continuous fields and a discrete topological sector. The underlying

fields include the scalar field θ ($f : \mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$), the vector field u ($f : \mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$), the emergent spinor ψ ($f : \mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$), and the filament current J ($f : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$). The continuous field action is **Fréchet-differentiable** over field space.

The Lagrangian density $\mathcal{L}_{\text{total}}$ includes standard field terms, the BF Topological Action (S_{BF}), and the filament kernel contribution:

$$\mathcal{L}_{\text{total}} = (\theta^\mu x)^2 + (u^\mu x)^2 + (J^{\mu\mu} x)^2 + S_{\text{BF}}(u, \theta, \tau) + \text{filament_kernel}(J, \tau) \quad (1)$$

The action S_{total} is the integral of $\mathcal{L}_{\text{total}}$. The topological action S_{BF} explicitly incorporates the discrete twist charge $Q(\tau)$. The discrete topological action (S_{Top}) is specified as σ -finite over the discrete twist ensemble (τ), ensuring a mathematically well-defined path integral.

B. Metric Variation and Emergent Gravity

Variation with respect to the emergent metric ($g_{\mu\nu}$), defined by the filament bundle frame field (E^I_μ), yields the Emergent Einstein Equation:

$$\frac{c^4}{8\pi G_{\text{ind}}} G_{\mu\nu}[g] = T_{\mu\nu}^{(u)} + T_{\mu\nu}^{(\theta)} + T_{\mu\nu}^{(\psi)} + \dots \quad (2)$$

The right-hand side is sourced by the stress-energy tensors (T) of the emergent fields (θ, u, ψ). The detailed derivation of this relation links filament micro-physics (tension T , length ℓ_f , strain κ , linking density ρ_G) to the macroscopic emergent constants via the **Micro-to-Macro Scaling Relations**. Applying the field equations to the spherically symmetric case is required to formally recover known fixed points, such as the **Schwarzschild metric**.

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III. EMERGENT QUANTUM MECHANISMS

A. Geometric Quantization

Intrinsic particle properties, specifically Spin and Flavor, are not axiomatic inputs but are derived from the geometric and topological attributes of the underlying filament helices. These quantum numbers are routinely recovered via the **Holonomy Quantization** mechanism. This requires calculating the **winding number** of the compact phase field (θ) around a closed loop defined on the manifold, which, multiplied by π , yields the quantized physical property.

B. Confinement and Gauge Structure

The topological structure provides a geometric basis for strong interaction phenomena, enforced by the emergent topological twist field (τ) through a $\mathbb{Z}_3$ fusion rule. This rule serves as a direct topological analogy to the three-quark structure required by Quantum Chromodynamics (QCD), accounting geometrically for the stability of three-constituent particles (baryons).

C. Projector Formalism

The theory employs the **spatial projector** ($\mathbf{P}_{\mu\nu}$), defined as $\mathbf{P}_{\mu\nu} = \mathbf{g}_{\mu\nu} + \mathbf{u}_\mu \mathbf{u}_\nu$. This projector restricts the dynamics to the three-dimensional subspace **orthogonal to** u^μ . This operation is necessary to eliminate unphysical longitudinal ghost modes and ensure that only the **two transverse polarizations** of the physical Temporon propagate.

IV. RESEARCH FRONTIERS

The pursuit of Level III completeness focuses on developing an informational story. This involves defining an entropy functional ($\mathbf{S}_{\text{filament}} = - \int \rho_\tau \log \rho_\tau \mathbf{d}\tau$) for the discrete twist ensemble (τ) to bridge the gap toward informational closure. The model includes an internal measurement structure, which attributes entanglement and collapse to worldline backbleed.

SUPPLEMENTAL MATERIAL

This material contains the detailed variational derivations and constraints required for full technical support of the main article.

APPENDIX A: DETAILED VARIATIONAL DERIVATIONS AND CONSERVATION LAWS

This section provides the formal variational derivations supporting the dynamical equations, demonstrating the rigorous consistency required by the Lean 4 verification check.

A.1. THE LAGRANGIAN DENSITY (NON-RELATIVISTIC LIMIT)

The non-relativistic fluid limit is governed by the dimensionally consistent Lagrangian density:

$$\mathcal{L} = \frac{1}{2}\rho \mathbf{v}^2 - \rho W(\Phi) - \frac{\kappa}{2}(\nabla\Phi)^2 + \lambda \rho \Phi (\nabla \cdot \mathbf{v}) \quad (3)$$

The system's stationarity conditions ($\delta S = 0$) are computed under arbitrary compactly-supported variations of the independent fields ($\rho, \mathbf{v}, \Phi$) constrained by the Lagrange multiplier (χ) enforcing mass continuity. Variation with respect to the velocity field ($\mathbf{v}$) yields the canonical momentum relation, and variation with respect to the density (ρ) yields the Bernoulli relation.

A.2. DISCRETE TOPOLOGICAL SECTOR

The discrete twist charge (τ) is defined by its topological invariants, where $Q(\tau)$ is the total topological charge:

$$Q(\tau) := \tau.L_{\text{wind}} + \tau.L_{\text{link}} + \tau.W_{\text{writhe}} \quad (4)$$

The topological mass suppression law depends inversely on this charge:

$$m_{\text{topo}}(m, \tau) := m / (Q\tau : \mathbb{R}) \quad (5)$$

The stability constraint, $\mathbf{Q}_{\text{stable}}(\tau) : \text{Prop} := \mathbf{Q}\tau \leq \mathbf{3}$, limits dynamically stable configurations in the theory.

APPENDIX B: GEOMETRIC ORIGINS OF PARTICLE PROPERTIES

B.1. HOLONOMY AND QUANTUM NUMBERS

Intrinsic particle properties are emergent properties derived from the geometry and topology of the underlying filament helices. The Holonomy Quantization mechanism involves calculating the winding number of the compact phase field (θ) around a closed loop defined on the manifold. The winding number multiplied by π yields the quantized physical property.

B.2. STRONG INTERACTION CONFINEMENT

The stability of three-constituent particles (baryons) is geometrically accounted for by the emergent topological twist field (τ) enforcing a $\mathbb{Z}_3$ **fusion rule**, serving as a direct topological analogy to the three-quark structure of QCD.

APPENDIX C: MASS SPECTRUM AND STABILITY

The mass spectrum follows a quadratic string formula for unstable excited states ($N \geq 1$):

$$m_{\text{string}}^2 = k \cdot (N + \beta \cdot Q^2 - a) \quad (6)$$

The consistency and stability of the particle definition rely on the **Topological Dominance constraint** ($\mathbf{R} \equiv \mathbf{m}_{\text{ind}}/\mathbf{m}_{\text{topo}} \ll 1$).

V. I. FORMAL CONSISTENCY AND RIGOR

The foundation of the Scalar-Angular-Torsion (SAT) theory relies on securing Level I completeness through rigorous geometric definitions and computational verification.

A. Mathematical Constraints on the Total Action

The comprehensive dynamics are governed by the Total Covariant Action (S_{SAT}), which integrates over continuous fields (ξ) and sums over stable discrete topological configurations (τ) via a generalized partition function (Z_{SAT}). To achieve a mathematically "Well-defined path integral," the hybrid action must satisfy two core requirements:

1. The action must be σ -finite over the discrete topological space (τ).
2. The action must be **Fréchet-differentiable** over the continuous field space (ξ).

These constraints define the geometric field space necessary for formal derivation.

B. Lean 4 Certification and Stability Proofs

The algebraic consistency and numerical predictions of the model are confirmed via **Lean 4 formal certification**. This process verifies the explicit variational equations, conservation laws, and dimensional constraints.

A critical stability condition, termed the ****UV Finiteness Lock****, emerges from these formal constraints, restricting dynamically stable filament configurations to a

maximum topological charge: $\mathbf{Q} \leq 3$. This lock is necessary to maintain:

- **Positive Effective Mass** ($\mathbf{m}_{\text{eff}} > 0$), ensuring particles possess observable mass.
- Adherence to the **Topological Dominance Condition** ($\mathbf{R} \equiv \mathbf{m}_{\text{ind}}/\mathbf{m}_{\text{topo}} \ll 1$), proving that the intrinsic topological mass dominates the induced inertial mass.

VI. II. DIMENSIONAL ANCHORING AND PREDICTED SPECTRUM

The framework operates with the constraint that **no numerical parameters are introduced axiomatically**. The entire physical scale is set by anchoring the effective mass of the $Q = 1$ topological state to an empirical value:

$$\mathbf{m}_{\text{eff}}(Q = 1) \approx 1.0073 \times 10^{-27} \text{ kg} \quad (7)$$

This normalization structurally fixes the dimensionless coupling ratio at $\mathbf{R} \approx 0.007304$, which is consistent with the empirical value of the fine-structure constant (α).

A. Topological Mass Suppression Law

The mass of a particle is governed by the Topological Mass Suppression Law, where mass is inversely proportional to the Topological Charge $Q(\tau)$, ensuring that greater topological complexity yields smaller mass.

B. Mass Spectrum Formula

The resulting mass spectrum is rigorously described by a quadratic string formula for excited states ($N \geq 1$):

$$m_{\text{string}}^2 = k \cdot (N + \beta \cdot Q^2 - a) \quad (8)$$

C. Novel Stable Ground States

The constraint imposed by the UV Finiteness Lock ($\mathbf{Q} \leq 3$) predicts the existence of two additional stable topological states (ground states, $N = 0$), serving as primary dark matter candidates:

- $\mathbf{Q} = 2$ state: $\approx 0.5073 \times 10^{-27} \text{ kg}$
- $\mathbf{Q} = 3$ state: $\approx 0.3403 \times 10^{-27} \text{ kg}$

VII. III. GEOMETRIC GAUGE MECHANISMS

A. Strong Interaction Confinement

The geometric basis for strong interaction confinement is achieved by the emergent topological twist field (τ) enforcing a $\mathbb{Z}_3$ fusion rule. This rule directly models the stability of three-constituent particles (baryons), providing a geometric equivalent to the requirements of Quantum Chromodynamics (QCD).

B. Quantum Number Emergence

Intrinsic quantum properties, such as Spin and Flavor, are recovered via the **Holonomy Quantization** mechanism. This requires calculating the winding number of the compact phase field (θ) around a closed loop on the manifold, which, when multiplied by π , yields the quantized physical property.

VIII. FIELD ALGEBRA AND DEFINING TENSORS

The continuous dynamics of the Scalar-Angular Torsion (SAT) framework are governed by a covariant geometric field model. The system operates on fields including the scalar field θ , the unit time-flow vector u , the emergent spinor ψ , and the filament current J [Manuscript Draft].

A. A. Field Definitions and Derivatives

The theory utilizes symbolic covariant derivatives (∇, D) on the emergent fields [Manuscript Draft]. The primary dynamic components that contribute to the generalized stress-energy tensor are the geometric Strain (S) and Stress (T) fields, defined based on these underlying field structures:

The ****Strain Field (S)**** is derived from the scalar (θ) and vector (u) fields:

$$S_{\mu\nu} = \nabla_\mu \theta_\nu + \nabla_\nu \theta_\mu + \frac{1}{2}(u_\mu \nabla_\nu \theta - u_\nu \nabla_\mu \theta) \quad (9)$$

The ****Stress Field (T)**** is defined by the interaction of the scalar field with the derivative of the vector field:

$$T^{\mu\nu} = \rho(\nabla^\mu u^\nu + u^\mu \nabla^\nu \theta) + J^\mu \theta^\nu \quad (10)$$

The **Filament Current (J)** is mathematically defined as a function of the discrete twist charge (τ) and continuous spacetime position (x): J is functionally written as $J^{\mu\mu}x$ within the core Lagrangian terms [Manuscript Draft, Variational Methods and Geometric Emergence in SAT Theory].

IX. II. HYDRODYNAMIC LIMIT AND CONSERVATION

The non-relativistic hydrodynamic limit provides the explicit spatial and temporal conservation laws governing the coarse-grained dynamics of the field model.

A. A. Non-Relativistic Lagrangian Density

The non-relativistic fluid limit is governed by the dimensionally consistent Lagrangian density ($\mathcal{L}$), which ensures stationarity conditions ($\delta S = 0$) can be computed under arbitrary compactly-supported variations of the independent fields ($\rho, \mathbf{v}, \Phi$):

$$\mathcal{L} = \frac{1}{2}\rho\mathbf{v}^2 - \rho W(\Phi) - \frac{\kappa}{2}(\nabla\Phi)^2 + \lambda\rho\Phi(\nabla\cdot\mathbf{v}) \quad (11)$$

Variation with respect to the density (ρ) yields the Bernoulli relation, and variation with respect to the velocity field ($\mathbf{v}$) yields the canonical momentum relation [Variational Methods and Geometric Emergence in SAT Theory].

B. B. Momentum Balance Equation

The system must satisfy conservation laws, yielding the momentum balance equation (Euler-Lagrange system for the fluid elements) [Lean-4 verified geometric constraint yields charge-dependent mass spectrum]:

$$\rho(\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v}) = -\nabla P - \rho \nabla W(\Phi) - \kappa(\nabla^2 \Phi) \nabla \Phi + \lambda \nabla(\rho \Phi) \quad (12)$$

X. III. THEORETICAL AND INFORMATIONAL CLOSURE

The project explicitly aims for high-level theoretical completion, necessitating constraints beyond empirical matching.

A. A. Completeness Levels and Semantic Unity

The framework attempts to synthesize physics by rigorously formalizing predictions derived exclusively from the action principle and variational equations [Variational Methods and Geometric Emergent in SAT Theory]. This rigor extends to formal verification using the Lean 4 computational environment [Lean-4 verified geometric constraint yields charge-dependent mass spectrum].

The demonstration that the physics can be formally verified in Lean 4 constitutes an "embryonic gesture toward **semantic unity**," implying that physics is, in

principle, reconstructible within a proof-assistant logic [Scalar-Angular Torsion: Unified Geometric Physics] [Manuscript Draft]. This level of completeness is referred to as **Metaphysical Closure Gesture (Level IV)** [Scalar-Angular Torsion: Unified Geometric Physics].

B. B. Informational Closure and Entanglement

To pursue Level III completeness ("informational story"), the model introduces specific structures related to the discrete nature of the field system [Variational Methods and Geometric Emergence in SAT Theory].

1. **Entropy Functional:** Informational closure requires defining an entropy functional for the discrete twist ensemble (τ), the collection of stable topological configurations:

$$S_{\text{filament}} = - \int \rho_{\tau} \log \rho_{\tau} d\tau \quad (13)$$

where ρ_{τ} is the density of the twist ensemble [Variational Methods and Geometric Emergence in SAT Theory] [Lean-4 verified geometric constraint yields charge-dependent mass spectrum]. 2. **Quantum Phenomena:** The model attributes entanglement and wave-function collapse to **worldline backbleed** [Variational Methods and Geometric Emergence in SAT Theory].

C. C. Aesthetic and Topological Constraint

The commitment to **Zero-Parameter Economy** is integral to the theory's claim of aesthetic closure, serving as evidence of mathematical inevitability [Scalar-Angular Torsion: Unified Geometric Physics] [Manuscript Draft]. All dimensionless constants are required to derive solely from internal geometry and topology [Manuscript Draft].

XI. IV. EMERGENCE OF GRAVITATION AND QUANTUM STRUCTURE

A. A. GEOMETRIC FOUNDATIONS OF UNIFICATION

The Scalar-Angular Torsion (SAT) framework achieves unification structurally at the ****intersection of the filament with the timesheet****. The theory encompasses General Relativity (GR), the Standard Model (SM), Quantum Field Theory (QFT), Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD), and recovers part of String Theory (ST). The formal realization of this structure defines the standard gauge group ($\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$) as a ****fiber bundle over the emergent metric manifold****.

The model is defined as a **mathematical model without ontological commitment** [Variational Methods and Geometric Emergence in SAT Theory].

B. B. EMERGENT GRAVITATIONAL DYNAMICS

Gravitation is recreated by the collective ****back-pull**** exerted by the filament structure on the time surface^{**}, which generates curvature. The variational principle, applied to the emergent metric, leads to the ****Emergent Einstein Equation**** [Variational Methods and Geometric Emergence in SAT Theory].

The theory establishes ****Micro-to-Macro Scaling Relations****, which link filament microphysics (tension T , length ℓ_f , strain κ , linking density ρ_G) to macroscopic emergent constants [Variational Methods and Geometric Emergence in SAT Theory].

The induced gravitational constant (G_{ind}) is shown to scale geometrically as:

$$G_{\text{ind}} \sim \frac{c^4 \ell_f^2}{16\pi \mathcal{A} \kappa} \quad (14)$$

Consistency validation requires that the numerical output derived from these parameters must match the known empirical value:

C. C. HOLONOMY QUANTIZATION AND CONFINEMENT

Intrinsic quantum properties, such as Spin and Flavor, are derived geometrically via the ****Holonomy Quantization**** mechanism [432, Variational Methods and Geometric Emergence in SAT Theory].

This mechanism involves calculating the ****winding number**** of the compact phase field (θ) around a closed loop defined on the manifold. This winding number, multiplied by π , yields the quantized physical property [Variational Methods and Geometric Emergence in SAT Theory].

Particle stability in the strong interaction is accounted for by the emergent topological twist field (τ) enforcing a $\mathbb{Z}_3$ fusion rule. This rule serves as a direct topological analogy to the three-quark structure required by Quantum Chromodynamics (QCD) [432, Variational Methods and Geometric Emergence in SAT Theory].

D. D. STABILITY CONSTRAINTS AND PREDICTIONS

The stability of the system is governed by three fundamental, formally certified constraints:

1. **Positive Effective Mass ($m_{\text{eff}} > 0$).** 2. **Positive Topological Charge ($Q > 0$).** 3. **Topological Dominance ($R \equiv m_{\text{ind}}/m_{\text{topo}} \ll 1$).**

The adherence to these constraints, verified by ****Lean 4 formal certification****, imposes the ****UV Finiteness Lock****, limiting dynamically stable ground states to configurations with topological charges $Q \leq 3$. This constraint leads directly to the prediction of stable, low-mass

dark matter candidates at $Q = 2$ ($\approx 0.5073 \times 10^{-27}$ kg) and $Q = 3$ ($\approx 0.3403 \times 10^{-27}$ kg).

Unstable excited states ($N \geq 1$) are predicted to follow a quadratic mass-squared spectrum formula [427, Lean-4 Verified Mass Spectrum]:

$$\mathbf{m}^2 = \mathbf{k}(\mathbf{N} + \beta \mathbf{Q}^2 - \mathbf{a}) \quad (15)$$

The derived dimensionless coupling ratio, fixed by the $Q = 1$ mass anchor, is $\mathbf{R} \approx 0.007304$.

XII. V. AXIOMATIC STRUCTURE AND FORMAL VERIFICATION

A. A. Axiomatic Foundation and Ontology

The Scalar-Angular Torsion (SAT) theory is built upon a minimal geometric ontology consisting of three fundamental primitives: the **Filament** (the 4D particle worldline), the **Time Surface** (Σ_t , the advancing 3D wavefront), and the **Particle** (the localized point of intersection) [Manuscript Draft]. The theory aims to structurally integrate General Relativity (GR), the Standard Model (SM), Quantum Field Theory (QFT), Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD), and part of String Theory (ST) by mapping them onto the geometry of the timesheet/filament intersection.

B. B. Zero-Parameter Economy and Semantic Unity

A central constraint is the **Zero-Parameter Economy** (Normalization Principle), asserting that the theory introduces **zero numerical parameters** axiomatically into its fundamental structure [Manuscript Draft]. This constraint supports the claim of **aesthetic closure** and mathematical inevitability. The formal structure, verified via **Lean 4 formal certification**, implies that physics may be reconstructible within a proof-assistant logic, serving as an embryonic gesture toward **semantic unity** (Metaphysical Closure Gesture, Level IV) [461, Manuscript Draft].

XIII. VI. THE TOPOLOGICAL MASS MECHANISM

The theory replaces the Higgs mechanism by defining mass as an emergent property of geometric tension and topology [Manuscript Draft].

A. A. Mass Definition and Suppression Law

The total observable mass ($\mathbf{m}_{\text{eff}}$) is the sum of the dominant **Topological Mass** (m_{topo}) and the sec-

ondary **Induced Inertial Mass** (m_{ind}) [Manuscript Draft, 455]. The mass is fundamentally governed by the **Topological Mass Suppression Law**, which dictates an inverse relationship between topological mass and the discrete Topological Charge $Q(\tau)$:

$$m_{\text{topo}}(m, \tau) := m / (Q\tau : \mathbb{R}) \quad (16)$$

The Induced Mass (m_{ind}) arises from dynamic drag against bosonic excitations (Temporons), defined in the Lean 4 structure as:

$$m_{\text{ind}}(\text{params})(E_{\text{overlap}}) := \text{params}.g_1^2 / \text{params}.M^2 * E_{\text{overlap}} \quad (17)$$

B. B. Stability and the Mass Spectrum

Particle stability requires the **Topological Dominance Condition** ($\mathbf{R} \equiv \mathbf{m}_{\text{ind}} / \mathbf{m}_{\text{topo}} \ll 1$) to be satisfied [Manuscript Draft, 88, 459]. The mass spectrum for unstable excited states ($N \geq 1$) is defined by a quadratic string formula:

$$\mathbf{m}^2 = \mathbf{k}(\mathbf{N} + \beta \mathbf{Q}^2 - \mathbf{a}) \quad (18)$$

XIV. VII. QUANTITATIVE CALIBRATION AND CONSISTENCY

The predictive power of the framework relies on successfully converging the geometrically derived ratios onto empirical constants (Calibration P).

A. A. Empirical Anchors

The two critical empirical anchors successfully established during calibration are:

1. **Hadronic Mass Scale** ($Q = 1$): Final Effective Mass $\mathbf{m}_{\text{eff}} \approx 1.0073 \times 10^{-27}$ kg.
2. **Coupling Ratio** (R): Final dimensionless ratio $\mathbf{R} \approx 0.007304$, which converges onto the known empirical value of the fine-structure constant ($\alpha \approx 0.007297$).

B. B. Cross-Sector Consistency Validation

The consistency of the calibrated parameters was verified across disparate sectors of physics:

- **Electroweak Check**: The calibrated coupling ratio ($R \approx 0.007304$) successfully facilitates the calculation of the correct W-to-Z boson mass ratio (target empirical value: **0.88145(13)**).
- **Gravitational Check**: The derived numerical value for the induced gravitational constant (G) must match the known empirical value:

XV. VIII. FALSIFIABILITY AND DARK MATTER PREDICTIONS

A. A. UV Finiteness Lock

The **Lean 4** formal verification confirms the structural constraint imposed by the **UV Finiteness Lock**, which mandates that only particle configurations with a Topological Charge $Q \leq 3$ can be dynamically stable ground states in 4D spacetime. This lock is necessary to meet the **Positive Effective Mass** and **Topological Dominance Condition** requirements.

B. B. Novel Stable States

Based on the $Q \leq 3$ constraint, the framework structurally predicts two novel stable topological states ($N = 0$) proposed as primary candidates for dark matter:

- $Q = 2$ state: $\approx 0.5073 \times 10^{-27}$ kg.
- $Q = 3$ state: $\approx 0.3403 \times 10^{-27}$ kg.

The successful detection of these specific, stable, low-mass particles is presented as the primary test of the framework.

XVI. VIII. EMPIRICAL VALIDATION AND DARK MATTER PREDICTIONS

A. A. ULTIMATE EMPIRICAL VALIDATION CHECK

The progression of the Scalar-Angular-Torsion (SAT) theory from self-consistent theoretical prediction to direct test against physical observation hinges on the **Ultimate Empirical Validation Check**. This check centers on the dedicated experimental search for the novel stable states constrained by the **UV Finiteness Lock** ($Q \leq 3$).

B. B. NOVEL STABLE PARTICLE SEARCH TARGETS

The dimensional anchor utilized in the theory fixed the $Q = 1$ state to the hadronic mass scale ($\approx 1.0073 \times 10^{-27}$ kg). The search program must target the precise effective masses ($N = 0$ ground states) predicted for the two remaining topologically allowed stable configurations:

1. **The $Q = 2$ state** is predicted to have an effective mass of $\approx 0.5073 \times 10^{-27}$ kg.
2. **The $Q = 3$ state** is predicted to have an effective mass of $\approx 0.3403 \times 10^{-27}$ kg.

C. C. SIGNIFICANCE: DARK MATTER CANDIDACY

The predicted mass and stability of the $Q = 2$ and $Q = 3$ states position them as **leading candidates** for dark matter.

1. **Stability:** Their stability is guaranteed because they are $N = 0$ ground states constrained by topological charge Q .
2. **Decoupling:** For dark matter candidacy, these particles must possess **suppressed electroweak interactions**. This requirement is mathematically verified if they satisfy the **Topological Dominance Condition** ($R \equiv m_{\text{ind}}/m_{\text{topo}} \ll 1$) [3, 459, Manuscript Draft].
3. **Gravitational Interaction:** Since they possess non-zero mass, they are confirmed to contribute to gravitational curvature via the geometric **“back-pull”** mechanism.

XVII. IX. COMPLETION OF GEOMETRIC PARTICLE IDENTITY

The final structural step in defining particle identity involves applying the **Holonomy Quantization** mechanism to these stable states to derive their intrinsic quantum numbers.

A. A. Holonomy Quantization

This process involves calculating the **winding number** of the compact phase field (θ) around a closed loop, which yields intrinsic properties like **Spin and Flavor** [3, Variational Methods and Geometric Emergence in SAT Theory]. The derivation confirms the geometric origin of spin based on filament structure:

- **Fermions** are identified as the structural filaments themselves and possess **half-integer spin**.
- **Bosons** are defined as parasitic ripples propagating along the fermionic substrate (filaments) and possess **integer spin** [4, Manuscript Draft].

XVIII. VII. COVARIANT FORMULATION AND TOPOLOGICAL ALGEBRA

This Supplemental Material details the formal covariant action and the algebraic structures governing the coupling of continuous fields (ξ) to the discrete topological configurations (τ) in the Scalar-Angular Torsion (SAT) framework.

A. A. Generalized Partition Function

The comprehensive dynamics are governed by the ****Total Action** (S_{SAT}), which is expressed via the generalized partition function Z_{SAT} . This functional integrates over continuous fields and sums over stable discrete topological configurations:

$$Z_{\text{SAT}} = \sum_{\tau} \int \mathcal{D}[\xi] \exp \left(\frac{i}{\hbar} \mathbf{S}_{\text{total}}[\xi, \tau] \right) \quad (19)$$

The action S_{total} must be σ -finite over the discrete twist ensemble (τ) and ****Fréchet-differentiable**** over the continuous field space (ξ) [Variational Methods and Geometric Emergence in SAT Theory].

B. B. Topological Charge $\mathbb{Z}_3$ Projection

The discrete twist charge (τ) is central to mass emergence and geometric gauge structure [Manuscript Draft]. The components of the structure ‘TopoInv’ (winding number L_{wind} , linking number L_{link} , and writhe number W_{writhe}) are constrained by a $\mathbb{Z}_3$ fusion rule, essential for modeling the Strong Interaction [402, Variational Methods and Geometric Emergence in SAT Theory]. This constraint is implemented algebraically by projecting the integers ($\mathbb{N}$) onto the finite quotient set $\mathbb{Z}_3$:

$$\begin{aligned} \tau_{\text{proj}} &= \{L_{\text{wind}} := \tau.L_{\text{wind}} \pmod{3}, \\ L_{\text{link}} &:= \tau.L_{\text{link}} \pmod{3}, \\ W_{\text{writhe}} &:= \tau.W_{\text{writhe}} \pmod{3}\} \end{aligned} \quad (20)$$

The total Topological Charge $Q(\tau)$ is the sum of these invariants [Manuscript Draft].

C. C. Component Actions and Field Coupling

The total Lagrangian density (L_{total}) incorporates fields (θ, u, J) and parameters into the action calculation [402, Variational Methods and Geometric Emergence in SAT Theory].

1. 1. BF Topological Action

The topological coupling is made explicit in the BF Topological Action (S_{BF}), which couples fields B and A to the real representation of the topological charge $Q(\tau)$:

$$S_{\text{BF}}(B, A, \tau) = \int_0^1 x, (Bx) \cdot (\nabla A x) + (Q(\tau) : \mathbb{R}) \quad (21)$$

(Note: The expression from the source is transcribed using standard integral notation and vector/scalar fields).

2. 2. Filament Kernel Contribution

The coupling defined by the ‘filament_kernel’ is responsible for suppressing topological mass (m_{topo}) through an exponential dependence on $Q(\tau)$:

$$\text{filament_kernel}(J, \tau) = \int_0^1 x, Jx \cdot \exp(-(Q(\tau) : \mathbb{R})) \quad (22)$$

The total Lagrangian density (L_{total}) combines these continuous and topological terms.

XIX. VIII. EXCITED STATE MASS PREDICTIONS

The stability constraint imposed by the ****UV Finiteness Lock**** requires stable ground states ($N = 0$) to satisfy $\mathbf{Q} \leq \mathbf{3}$ [SUPP 5, SUPP 4]. The mass of unstable ****excited states** ($N \geq 1$) is governed by a quadratic spectrum formula [404, SUPP 4]:

$$\mathbf{m}^2 = \mathbf{k}(\mathbf{N} + \beta \mathbf{Q}^2 - \mathbf{a}) \quad (23)$$

Applying this spectrum to the first excited state ($N = 1$) yields specific mass-squared predictions for the topologically allowed charges:

$$Q = 1, N = 1 \rightarrow \mathbf{1.1} \quad (24)$$

$$Q = 2, N = 1 \rightarrow \mathbf{1.4} \quad (25)$$

$$Q = 3, N = 1 \rightarrow \mathbf{1.9} \quad (26)$$

(These predicted values are relative mass-squared units derived from the internal geometry, analogous to normalized calibration points).

A. A. Geometric Unification Scope

The framework is claimed to cover ****GR, SM, QFT, QED, QCD, and recovers part of ST****,

with all sectors unified at the intersection of the filament with the timesheet. The geometry of unification defines the standard gauge group ($\mathbf{SU(3)} \times \mathbf{SU(2)} \times \mathbf{U(1)}$) as a **fiber bundle over the emergent metric manifold** [SUPP 4].

XX. IX. GEOMETRIC UNIFICATION ARCHITECTURE

A. A. Scope of Synthesis

The Scalar-Angular Torsion (SAT) theory is a geometric framework covering **GR, SM, QFT, QED, QCD**, and recovers part of **ST**. Unification is achieved structurally at the geometric **intersection of the filament with the timesheet** [341, SUPP 7]. This geometric realization defines the standard gauge group ($\mathbf{SU(3)} \times \mathbf{SU(2)} \times \mathbf{U(1)}$) as a **fiber bundle over the emergent metric manifold** [SUPP 4, SUPP 7].

B. B. Foundational Primitives

The minimal geometric ontology is defined by the **Filament** (4D particle worldline), the **Time Surface** (Σ_t , the advancing 3D wavefront), and the **Particle** (the localized point of intersection) [Manuscript Draft]. The entire framework is advanced as a **mathematical model without ontological commitment** [Variational Methods and Geometric Emergence in SAT Theory]. The curvature observed in General Relativity (GR) is attributed geometrically to the collective “back-pull” exerted by the filament structure on the time surface [Manuscript Draft].

XXI. X. FORMAL LOGIC AND VERIFICATION CONSTRAINTS

A. A. Formal Constraints and Stability Indicators

The numerical predictions and internal algebraic consistency rely upon formal certification using the Lean 4 computational environment [340, Lean 4 Verified Mass Spec...]. This process validates explicit variational equations and enforces three primary stability criteria:

1. **Positive Effective Mass** ($m_{\text{eff}} > 0$).
2. **Topological Dominance** ($R < 1$), requiring that the topological mass component outweighs the induced inertial mass component.

3. **Positive Topological Charge** ($Q > 0$), ensuring only non-trivial topological configurations are permitted.

The successful detection of the predicted stable particles ($Q = 2, 3$) serves as the primary test for the geometric foundation of the search program.

B. B. Constraint on Stability (UV Finiteness Lock)

The consistency verified by Lean 4 imposes the **UV Finiteness Lock**, restricting dynamically stable ground states to configurations with topological charges $Q \leq 3$ [SUPP 4, SUPP 5]. Adherence to the **Zero-Parameter Economy** is the framework’s ultimate honesty test, demanding that all dimensionless constants derive solely from internal geometry [Manuscript Draft].

XXII. XI. PARTICLE IDENTITY AND FIELD OPERATORS

A. A. Geometric Identity

Particle identity is determined by the geometrical structure and dynamics relative to the filament substrate:

- **Fermions** are identified as the structural **filaments** themselves, carrying helices that yield half-integer spin [Manuscript Draft, SUPP 6].
- **Bosons** (including Temporons) are defined as **dynamic torsional patterns** or **parasitic ripples** propagating along the fermionic substrate, and are **not independent filaments** [Manuscript Draft].

B. B. Spatial Projector Formalism

The continuous dynamics are constrained using a projection operator to ensure physical realism by managing kinematic degrees of freedom [Variational Methods and Geometric Emergence in SAT Theory]:

The **spatial projector** ($\mathbf{P}_{\mu\nu}$) is defined as:

$$\mathbf{P}_{\mu\nu} = \mathbf{g}_{\mu\nu} + \mathbf{u}_\mu \mathbf{u}_\nu \quad (27)$$

This projector restricts the dynamics to the three-dimensional subspace orthogonal to the unit time-flow vector (u^μ), which eliminates unphysical longitudinal ghost modes and ensures that only the

two transverse polarizations of the physical Temporon propagate [Variational Methods and Geometric Emergence in SAT Theory].

— **Analogy for Projector Formalism:* Think of the ***spatial projector*** as a camera filter used by the universe. It only allows ripples (Bosons) moving **across** the current of time (Filament flow, u^μ) to be recorded as real physical forces, while ignoring any energy trying to move **along** the flow of time that would violate stability or consistency rules. This keeps the cinematic physics clean by cutting out the unphysical special effects (ghost modes).

XXIII. XII. MODELING EXCITATION DYNAMICS AND UNSTABLE STATES

This Supplemental Material details the companion predictive capability of the Scalar-Angular Torsion (SAT) framework concerning dynamically unstable, excited particle states ($N \geq 1$).

A. A. Foundation of the Stable State Search Program

The ultimate empirical validation phase relies on the search for the ***novel stable states*** ($N = 0$ ground states) constrained by the topological algebra. The geometric foundation for this search program is fully defined, confirming the targets for detection:

- $\mathbf{Q} = 2$ state at $\approx 0.5073 \times 10^{-27}$ kg.
- $\mathbf{Q} = 3$ state at $\approx 0.3403 \times 10^{-27}$ kg.

For these particles, the necessary identity constraints (Spin/Flavor) are confirmed to be derivable via ***Holonomy Quantization***.

B. B. Predictive Capability: Unstable Particle Dynamics

The SAT theory extends beyond stable ground states to model ***Excitation Dynamics*** for unstable configurations, defined by the excitation quantum number $\mathbf{N} \geq 1$.

1. ***Governing Formula:*** The modeling of these unstable particles utilizes the ***quadratic mass spectrum formula***, which was previously introduced as:

$$\mathbf{m}^2 = k(\mathbf{N} + \beta \mathbf{Q}^2 - a) \quad [\text{SUPP 4, SUPP 5, SUPP 7}] \quad (28)$$

2. ***Dynamic Geometry:*** By applying this formula, the theory calculates the ***dynamic geometry of unstable particles***. This geometry differs significantly from the statically stable configurations constrained by the UV Finiteness Lock.

3. ***Decay Channel Foundation:*** The ability to model the dynamic geometry of unstable states is essential for establishing the foundation required for ***predicting their decay channels***. The decay is presumed to occur through topological transformations that relax the excited state ($N \geq 1$) down to one of the allowed stable ground states ($N = 0, Q \leq 3$) or by fragmenting into lower- Q stable components, consistent with conservation laws derived from the variational principle [Variational Methods and Geometric Emergence in SAT Theory].

XXIV. XIII. COUPLING RATIO CONSISTENCY

The robustness of the entire mass spectrum calculation, including both stable and unstable states, relies on the success of the Zero-Parameter Economy, which fixed the dimensionless coupling ratio at $\mathbf{R} \approx 0.007304$ [SUPP 5]. This value must be maintained consistently across the $N \geq 1$ calculations to ensure the predicted masses are dimensionally consistent.

XXV. XIV. MODELING EXCITATION DYNAMICS AND DECAY PREDICTION

This Supplemental Material details the mechanism within the Scalar-Angular-Torsion (SAT) framework used to predict the dynamics and stability of both ground states ($\mathbf{N} = 0$) and dynamically unstable excited particle states ($\mathbf{N} \geq 1$). This predictive capability relies fundamentally on the String/Filament Dynamics component embedded within the ***Total Covariant Action*** (S_{SAT}).

A. A. THE QUADRATIC MASS SPECTRUM FORMULA

Excited particle states ($N \geq 1$) are governed by the same quadratic mass spectrum formula utilized for ground state stability checks:

$$m_{\text{string}}^2 = k \cdot (N + \beta \cdot Q^2 - a) \quad (29)$$

The predicted mass-squared values for the first excited oscillator level ($\mathbf{N} = 1$) for the stable topological charges ($\mathbf{Q} \leq 3$) are confirmed to be:

- $\mathbf{Q} = \mathbf{1}, \mathbf{N} = \mathbf{1} \rightarrow \mathbf{1.1}$
- $\mathbf{Q} = \mathbf{2}, \mathbf{N} = \mathbf{1} \rightarrow \mathbf{1.4}$
- $\mathbf{Q} = \mathbf{3}, \mathbf{N} = \mathbf{1} \rightarrow \mathbf{1.9}$

B. FILAMENT DISPLACEMENT AND DECAY CHANNELS

The geometry of the filament bundle dictates the likelihood and nature of particle decay. The decay channels and interaction cross-sections for unstable resonances ($\mathbf{N} \geq \mathbf{1}$) are determined by modeling the filament's **transverse displacement** ($\mathbf{X}$ and $\mathbf{Y}$). This displacement is calculated based on the filament's specific list of **String Modes**:

$$\mathbf{X}(z, t) = \sum_n \alpha_x \sin\left(\frac{n\pi z}{L}\right) \cos(n\omega_0 t + \phi_x) \quad (30)$$

$$\mathbf{Y}(z, t) = \sum_n \alpha_y \sin\left(\frac{n\pi z}{L}\right) \cos(n\omega_0 t + \phi_y) \quad (31)$$

Here, z represents the position along the filament, t is time, n is the oscillator mode number, L is the length, ω_0 is the fundamental frequency, and the coefficients (α) and phases (ϕ) define the dynamic geometry of the excitation.

This modeling task predicts the **dynamic geometry of excited particles**, which is the necessary precursor for predicting **decay channels** and **interaction cross-sections** for these resonances.

C. STRUCTURAL CONSTRAINT ON HADRONIC STABILITY

The consistency of this modeling approach with the Strong Interaction sector requires that the underlying topological structure enforces the $\mathbb{Z}_3$ fusion rule. This rule provides the geometric reason for the stability of three-constituent particles (baryons), reinforcing consistency with Quantum Chromodynamics (QCD).

D. CONTEXT: STABLE STATE FOUNDATION

This dynamic modeling complements the ultimate empirical validation targets for the **novel stable states** ($\mathbf{N} = \mathbf{0}$), which are fully defined at $\mathbf{Q} = \mathbf{2}$ ($\approx 0.5073 \times 10^{-27}$ kg) and $\mathbf{Q} = \mathbf{3}$ ($\approx 0.3403 \times 10^{-27}$ kg). The necessary particle identity (Spin/Flavor) for these ground states is confirmed to be derivable via **Holonomy Quantization**.